

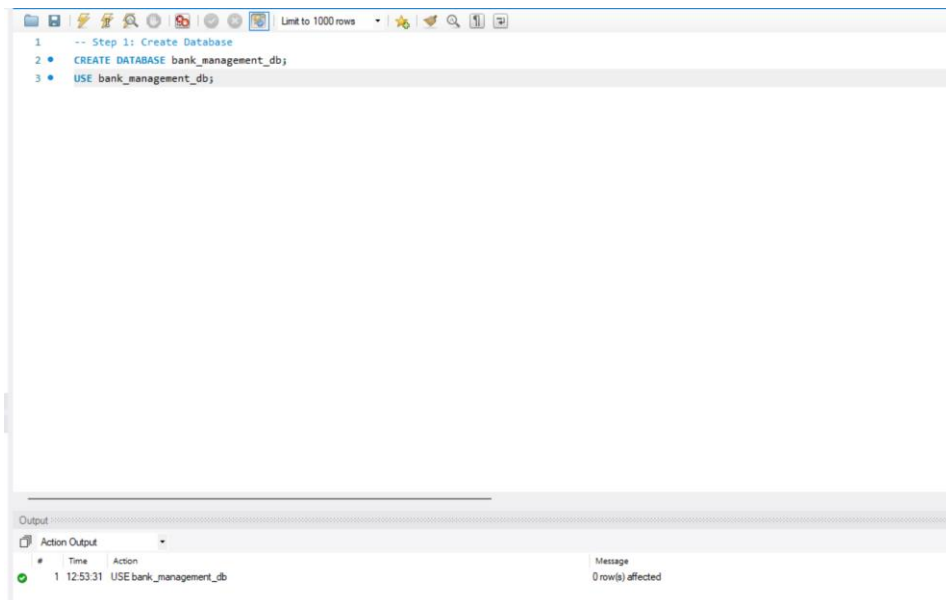
Commands

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Question :

Perform the following DDL, DML, DCL, and TCL operations on a Bank Management System database: create and alter a bank transaction table, add and modify columns, rename, truncate and drop tables; insert, update, delete, and retrieve transaction records using SELECT, WHERE, and ORDER BY; grant and revoke user privileges; and perform transaction control operations using COMMIT, SAVEPOINT, and ROLLBACK.

Step 1 — Create Database



Step 2 — DDL Commands

```

1  -- Step 1: Create Database
2  • CREATE DATABASE bank_management_db;
3  • USE bank_management_db;
4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );

```

Output

Action Output

#	Time	Action	Message
1	12:53:31	USE bank_management_db	0 row(s) affected
2	12:55:15	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected

```

1  -- Step 1: Create Database
2  • CREATE DATABASE bank_management_db;
3  • USE bank_management_db;
4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15 ADD account_no VARCHAR(20);

```

Output

Action Output

#	Time	Action	Message
1	12:53:31	USE bank_management_db	0 row(s) affected
2	12:55:15	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
3	12:56:01	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

```

1  -- Step 1: Create Database
2  • CREATE DATABASE bank_management_db;
3  • USE bank_management_db;
4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15   ADD account_no VARCHAR(20);
16 -- Step 4: Modify Customer Name Column
17 • ALTER TABLE bank_transactions
18   MODIFY customer_name VARCHAR(100);

```

Output

#	Time	Action	Message
✓ 1	12:53:31	USE bank_management_db	0 row(s) affected
✓ 2	12:55:15	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name VARCHAR(50), transaction_type VARCHAR(20), amount DECIMAL(10,2), transaction_date DATE)	0 row(s) affected
✓ 3	12:56:01	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 4	12:57:01	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

```

13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15   ADD account_no VARCHAR(20);
16 -- Step 4: Modify Customer Name Column
17 • ALTER TABLE bank_transactions
18   MODIFY customer_name VARCHAR(100);
19 -- Step 5: Rename Transaction Table
20 • RENAME TABLE bank_transactions
21   TO customer_transactions;
22

```

Output

#	Time	Action	Message
✓ 1	12:53:31	USE bank_management_db	0 row(s) affected
✓ 2	12:55:15	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name VARCHAR(50), transaction_type VARCHAR(20), amount DECIMAL(10,2), transaction_date DATE)	0 row(s) affected
✓ 3	12:56:01	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 4	12:57:01	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	12:57:58	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected

```

13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15   ADD account_no VARCHAR(20);
16 -- Step 4: Modify Customer Name Column
17 • ALTER TABLE bank_transactions
18   MODIFY customer_name VARCHAR(100);
19 -- Step 5: Rename Transaction Table
20 • RENAME TABLE bank_transactions
21   TO customer_transactions;
22 -- Step 6: Truncate Transaction Table
23 • TRUNCATE TABLE customer_transactions;
24

```

Output			
Action Output			
#	Time	Action	Message
✓ 1	12:53:31	USE bank_management_db	0 row(s) affected
✓ 2	12:55:15	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
✓ 3	12:56:01	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 4	12:57:01	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	12:57:58	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 6	13:04:15	TRUNCATE TABLE customer_transactions	0 row(s) affected

```

20 • RENAME TABLE bank_transactions
21   TO customer_transactions;
22 -- Step 6: Truncate Transaction Table
23 • TRUNCATE TABLE customer_transactions;
24 -- Step 7: Create Backup Table
25 • CREATE TABLE bank_backup(
26   txn_id INT,
27   customer_name VARCHAR(100)
28 );

```

Output			
Action Output			
#	Time	Action	Message
✓ 2	12:55:15	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	0 row(s) affected
✓ 3	12:56:01	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 4	12:57:01	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	12:57:58	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 6	13:04:15	TRUNCATE TABLE customer_transactions	0 row(s) affected
✓ 7	13:05:13	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected

```

24 -- Step 7: Create Backup Table
25 • CREATE TABLE bank_backup(
26   txn_id INT,
27   customer_name VARCHAR(100)
28 );
29 -- Step 8: Drop Backup Table
30 • DROP TABLE bank_backup;

```

Output			
Action Output			
#	Time	Action	Message
✓ 3	12:56:01	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 4	12:57:01	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 5	12:57:58	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 6	13:04:15	TRUNCATE TABLE customer_transactions	0 row(s) affected
✓ 7	13:05:13	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
✓ 8	13:06:40	DROP TABLE bank_backup	0 row(s) affected

Step 3: DML Commands

```

31 -- Step 9: Insert Sample Data
32 • INSERT INTO customer_transactions
33 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
34 VALUES
35 (101,'Nikshitha','Hyderabad','Deposit',5000,'2024-01-05'),
36 (102,'Bhanu priya','Hyderabad','Withdrawal',2000,'2024-01-06'),
37 (103,'Harshika','Vijayawada','Deposit',12000,'2024-01-08'),
38 (104,'Sathwika','Vizag','Deposit',8000,'2024-01-10'),
39 (105,'Archana','Hyderabad','Withdrawal',3500,'2024-01-11'),
40 (106,'Rishika','Vizag','Deposit',15000,'2024-01-12'),
41 (107,'Rithvika','Vijayawada','Withdrawal',1000,'2024-01-13'),
42 (108,'Srikanth','Hyderabad','Deposit',9000,'2024-01-14'),
43 (109,'Varsha','Vizag','Withdrawal',4000,'2024-01-15'),
44 (110,'Srija','Vijayawada','Deposit',11000,'2024-01-16');

```

Output

Action Output

#	Time	Action	Message
4	12:57:01	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	12:57:58	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
6	13:04:15	TRUNCATE TABLE customer_transactions	0 row(s) affected
7	13:05:13	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
8	13:06:40	DROP TABLE bank_backup	0 row(s) affected
9	13:09:40	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

```

45 -- Step 10: Insert New Transaction
46 • INSERT INTO customer_transactions
47 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
48 VALUES
49 (111,'Venu','Vizag','Deposit',7000,'2024-01-18');

```

Output

Action Output

#	Time	Action	Message
5	12:57:58	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
6	13:04:15	TRUNCATE TABLE customer_transactions	0 row(s) affected
7	13:05:13	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
8	13:06:40	DROP TABLE bank_backup	0 row(s) affected
9	13:09:40	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
10	13:11:46	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	1 row(s) affected

```

47 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
48 VALUES
49 (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
50 -- Step 11: Update Transaction Amount
51 • UPDATE customer_transactions
52 SET amount = 5000
53 WHERE txn_id = 105;
54

```

Output

Action Output

#	Time	Action	Message
8	13:06:40	DROP TABLE bank_backup	0 row(s) affected
9	13:09:40	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
10	13:11:46	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	1 row(s) affected
11	13:12:54	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	Error Code: 1062. Duplicate entry '111' for key 'customer_transactions.PRIMARY'
12	13:13:53	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
13	13:14:07	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

```

45 -- Step 10: Insert New Transaction
46 • INSERT INTO customer_transactions
47 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
48 VALUES
49 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
50 -- Step 11: Update Transaction Amount
51 • UPDATE customer_transactions
52 SET amount = 5000
53 WHERE txn_id = 105;
54 -- Step 12: Delete Transaction
55 • DELETE FROM customer_transactions
56 WHERE txn_id = 111;

```

Output

Action Output

#	Time	Action	Message
9	13:09:40	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
10	13:11:46	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	1 row(s) affected
11	13:12:54	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	Error Code: 1062. Duplicate entry '111' for key 'customer_transactions.PRIMARY'
12	13:13:53	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
13	13:14:07	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0
14	13:15:10	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected

```

53 WHERE txn_id = 105;
54 -- Step 12: Delete Transaction
55 • DELETE FROM customer_transactions
56 WHERE txn_id = 111;
57 -- Step 13: Display All Transactions
58 • SELECT *
59 FROM customer_transactions;

```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Nikshitha	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Bhanu priya	Hyderabad	Deposit	2000.00	2024-01-06	NULL
103	Harshika	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Sathwik	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Archana	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Rishika	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Rithvika	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Srikanth	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Varsha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Srija	Vijayawada	Deposit	11000.00	2024-01-16	NULL

transactions 1 x

Output

Action Output

#	Time	Action	Message
10	13:11:46	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	1 row(s) affected
11	13:12:54	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	Error Code: 1062. Duplicate entry '111' for key 'customer_transactions.PRIMARY'
12	13:13:53	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
13	13:14:07	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0
14	13:15:10	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
15	13:16:18	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned

```

57 -- Step 13: Display All Transactions
58 • SELECT *
59 FROM customer_transactions;
60 -- Step 14: Display Deposit Transactions
61 • SELECT *
62 FROM customer_transactions
63 WHERE transaction_type = 'Deposit';

```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Nikshitha	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Harshika	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Sathwika	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Rishika	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Srikanth	Hyderabad	Deposit	9000.00	2024-01-14	NULL
110	Srija	Vijayawada	Deposit	11000.00	2024-01-16	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

customer_transactions 2 x

Output

#	Time	Action	Message
✓ 12	13:13:53	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 13	13:14:07	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0
✓ 14	13:15:10	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
✓ 15	13:16:18	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
✓ 16	13:17:03	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit'	6 row(s) returned
✓ 17	13:17:12	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned

```

63 WHERE transaction_type = 'Deposit';
64 -- Step 15: Sort Transactions by Amount
65 • SELECT *
66 FROM customer_transactions
67 ORDER BY amount DESC;

```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Rishika	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Harshika	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Srija	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Srikanth	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Sathwika	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Nikshitha	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Archana	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
109	Varsha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
102	Bhanu priya	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
107	Rithvika	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

customer_transactions 3 x

Output

#	Time	Action	Message
✓ 14	13:15:10	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
✓ 15	13:16:18	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
✓ 16	13:17:03	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit'	6 row(s) returned
✓ 17	13:17:12	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
✓ 18	13:23:48	CREATE USER 'BranchManager'@localhost IDENTIFIED BY '1234'	0 row(s) affected
✓ 19	13:25:24	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned

Step 4: DCL Commands


```

64 -- Step 15: Sort Transactions by Amount
65 • SELECT *
66 FROM customer_transactions
67 ORDER BY amount DESC;
68 -- Create Auditor1 User
69 • CREATE USER 'Auditor1'@'localhost'
70 IDENTIFIED BY '1234';
71
72 -- Create BranchManager User
73 • CREATE USER 'BranchManager'@'localhost'
74 IDENTIFIED BY '1234';
75

```

Output

#	Time	Action	Message
✓ 13	13:14:07	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0
✓ 14	13:15:10	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
✓ 15	13:16:18	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
✓ 16	13:17:03	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit'	6 row(s) returned
✓ 17	13:17:12	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
✓ 18	13:23:48	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected

Step 4: DCL Commands

```

73 -- Create BranchManager User
74 • CREATE USER 'BranchManager'@'localhost'
75 IDENTIFIED BY '1234';
76 -- Step 16: Grant SELECT Permission
77 • GRANT SELECT
78 ON customer_transactions
79 TO 'Auditor1'@'localhost';

```

Output

#	Time	Action	Message
✓ 18	13:23:48	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
✓ 19	13:25:24	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned
✗ 20	13:26:37	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT


```

76 -- Step 16: Grant SELECT Permission
77 • GRANT SELECT
78 ON customer_transactions
79 TO 'Auditor1'@'localhost';
80 -- Step 17: Grant All Privileges
81 • GRANT ALL
82 ON customer_transactions
83 TO 'BranchManager'@'localhost';

```

Output

#	Time	Action	Message
✓ 19	13:25:24	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned
✗ 20	13:26:37	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
✗ 21	13:27:00	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
✗ 22	13:27:19	CREATE DATABASE bank_management_db	Error Code: 1007. Can't create database 'bank_management_db'; database exists
✗ 23	13:27:21	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
✓ 24	13:28:07	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected


```

82 ON customer_transactions
83 TO 'BranchManager'@'localhost';
84 -- Step 18: Revoke SELECT Permission
85 • REVOKE SELECT
86 ON customer_transactions
87 FROM 'Auditor1'@'localhost';

```

Output

#	Time	Action	Message
21	13:27:00	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
22	13:27:19	CREATE DATABASE bank_management_db	Error Code: 1007. Can't create database 'bank_management_db': database exists
23	13:27:21	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
24	13:28:07	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected

```

88 -- Step 19: Revoke All Privileges
89 • REVOKE ALL PRIVILEGES
90 ON customer_transactions
91 FROM 'BranchManager'@'localhost';
92

```

Output

#	Time	Action	Message
22	13:27:19	CREATE DATABASE bank_management_db	Error Code: 1007. Can't create database 'bank_management_db': database exists
23	13:27:21	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
24	13:28:07	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
25	13:28:59	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'customer_trans.
26	13:29:03	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'customer_trans.
27	13:30:15	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected

Step 5: TCL Commands

```

92 -- Step 20: Commit Transaction
93
94 • UPDATE customer_transactions
95 SET amount = 2000
96 WHERE txn_id = 102;
97
98 • COMMIT;

```

Output

#	Time	Action	Message
24	13:28:07	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
25	13:28:59	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'customer_trans.
26	13:29:03	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'customer_trans.
27	13:30:15	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
28	13:32:07	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	Error Code: 1147. There is no such grant defined for user 'BranchManager' on host 'localhost' on table 'custom.
29	13:32:17	COMMIT	0 row(s) affected

```

94 • UPDATE customer_transactions
95 SET amount = 2000
96 WHERE txn_id = 102;
97
98 • COMMIT;
99 -- Step 21: Create Savepoint
100 • START TRANSACTION;
101
102 • SAVEPOINT Before_Update;

```

Output

#	Time	Action	Message
25	13:28:59	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'customer_trans.
26	13:29:03	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'customer_trans.
27	13:30:15	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
28	13:32:07	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	Error Code: 1147. There is no such grant defined for user 'BranchManager' on host 'localhost' on table 'custom.
29	13:32:17	COMMIT	0 row(s) affected
30	13:33:03	SAVEPOINT Before_Update	0 row(s) affected

```

103 -- Step 22: Rollback to Savepoint
104 • UPDATE customer_transactions
105   SET amount = 99999
106   WHERE txn_id = 102;
107
108 • ROLLBACK TO SAVEPOINT Before_Update;
109 -- Reset Transaction State
110 • ROLLBACK;
111
112 • SET autocommit = 1;

```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✗	26 13:29:03	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'customer_trans.
✓	27 13:30:15	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
✗	28 13:32:07	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	Error Code: 1147. There is no such grant defined for user 'BranchManager' on host 'localhost' on table 'custom.
✓	29 13:32:17	COMMIT	0 row(s) affected
✓	30 13:33:03	SAVEPOINT Before_Update	0 row(s) affected
✓	31 13:34:07	SET autocommit = 1	0 row(s) affected

```

103 -- Step 22: Rollback to Savepoint
104 • UPDATE customer_transactions
105   SET amt = 99999
106   WHERE txn_id = 102;
107
108 • ROLLBACK TO SAVEPOINT Before_Update;
109 -- Reset Transaction State
110 • ROLLBACK;
111
112 • SET autocommit = 1;
113

```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓	27 13:30:15	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
✗	28 13:32:07	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	Error Code: 1147. There is no such grant defined for user 'BranchManager' on host 'localhost' on table 'custom.
✓	29 13:32:17	COMMIT	0 row(s) affected
✓	30 13:33:03	SAVEPOINT Before_Update	0 row(s) affected
✓	31 13:34:07	SET autocommit = 1	0 row(s) affected
✓	32 13:35:23	SET autocommit = 1	0 row(s) affected